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News May 2018

OSET-NTP Trial 12 Results
Environment Technology Technical Team
Bach/Holiday Home Wastewater System Upgrade with AES

OSET-NTP Trial 12, Rotorua: Much better than Secondary Treatment achieved for tested AES-38 system

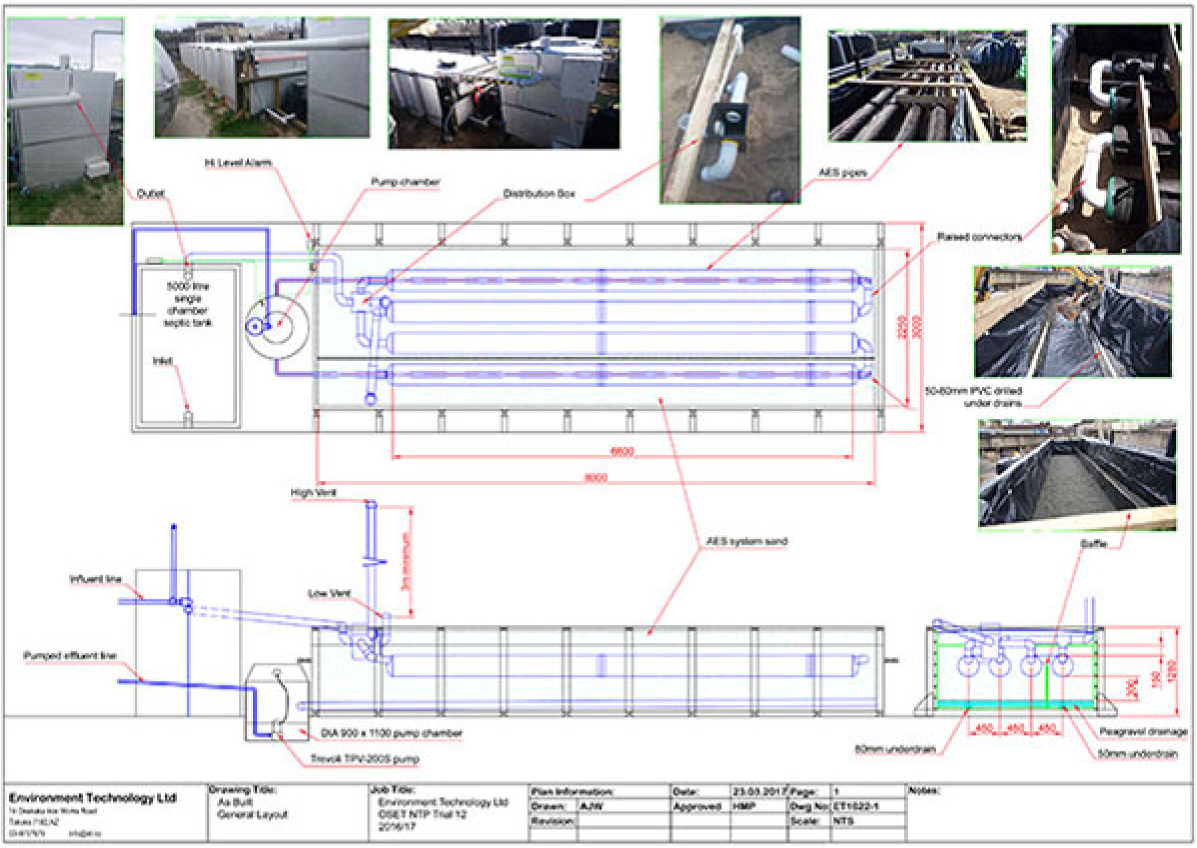
AES has received its Performance Certificate for the wastewater treatment system tested in OSET-NTP Trial 12. The results are shown below with BOD=2mg/L and TSS=3mg/. This easily achieves Advanced Secondary Treatment level which is frequently defined as BOD ≤10mg/L; TSS ≤10mg/L.

Indicator Parameters	Median	Std Dev	Rating	Rating System				
				A+	A	B	C	D
BOD (mg/L)	2	1.5	A+	<5	<10	<20	<30	≥30
TSS (mg/L)	3	3.1	A+	<5	<10	<20	<30	≥30
Total Nitrogen (mg/L)	37	4.9	D	<5	<15	<25	<30	≥30
NH_4- Nitrogen (mg/L)	1.8	0.5	A	<1	<5	<10	<20	≥20
Total phosphorus (mg/L)	2.7	0.5	B	<1	<2	<5	<7	≥7
Faecal Coliforms (cfu/100mL)	2,260	7,000	B	<10	<200	<10,000	<100,000	≥100,000
Energy (kWh/d) (mean)	0.1	0.04	A	0	<1	<2	<5	≥5

Environment Technology installed the AES-38 system at the Rotorua facility in September 2016. Trial 12 ran from October 2016 – July 2017; the AES-38 system operated without attendance throughout the trial. The full Performance Certificate can be downloaded from the ET website.

VIEW OSET-NTP CERTIFICATE   VIEW OSET-NTP NOTES

Below is the **As-Built plan**. You can also view photos of the installation on the ET website.



UV Filtration / Nitrogen Reduction in Trial 13

The AES-38 system at OSET-NTP for Trial 13, with the treated effluent being recirculated back through the anaerobic septic tank to further reduce total nitrogen (TN). The recirculation pump is on a timer, with gravity overflow through a Salcor 3G UV unit prior to the discharge tank. Results to date are replicating the TN results achieved in the Trial 12 research and development phase, where TN reduction of 82% was achieved.

Environment Technology Technical Team

Environment Technology's technical team are available to help with any questions you may have re suitability of an AES system for your site, council compliance, design or installation. We are experienced in small to large scale installations, can assist with design questions and plans, and work closely with many independent design companies and engineers around New Zealand.

Stan van Uden has recently joined Dick Lamb and Hazel Pearson on the technical team. Call us on 03 970 7979 / 0800 927 834 or email us.

Dick Lamb

Hazel Pearson

Stan van Uden

Bach/Holiday Home Wastewater Treatment Refit with AES

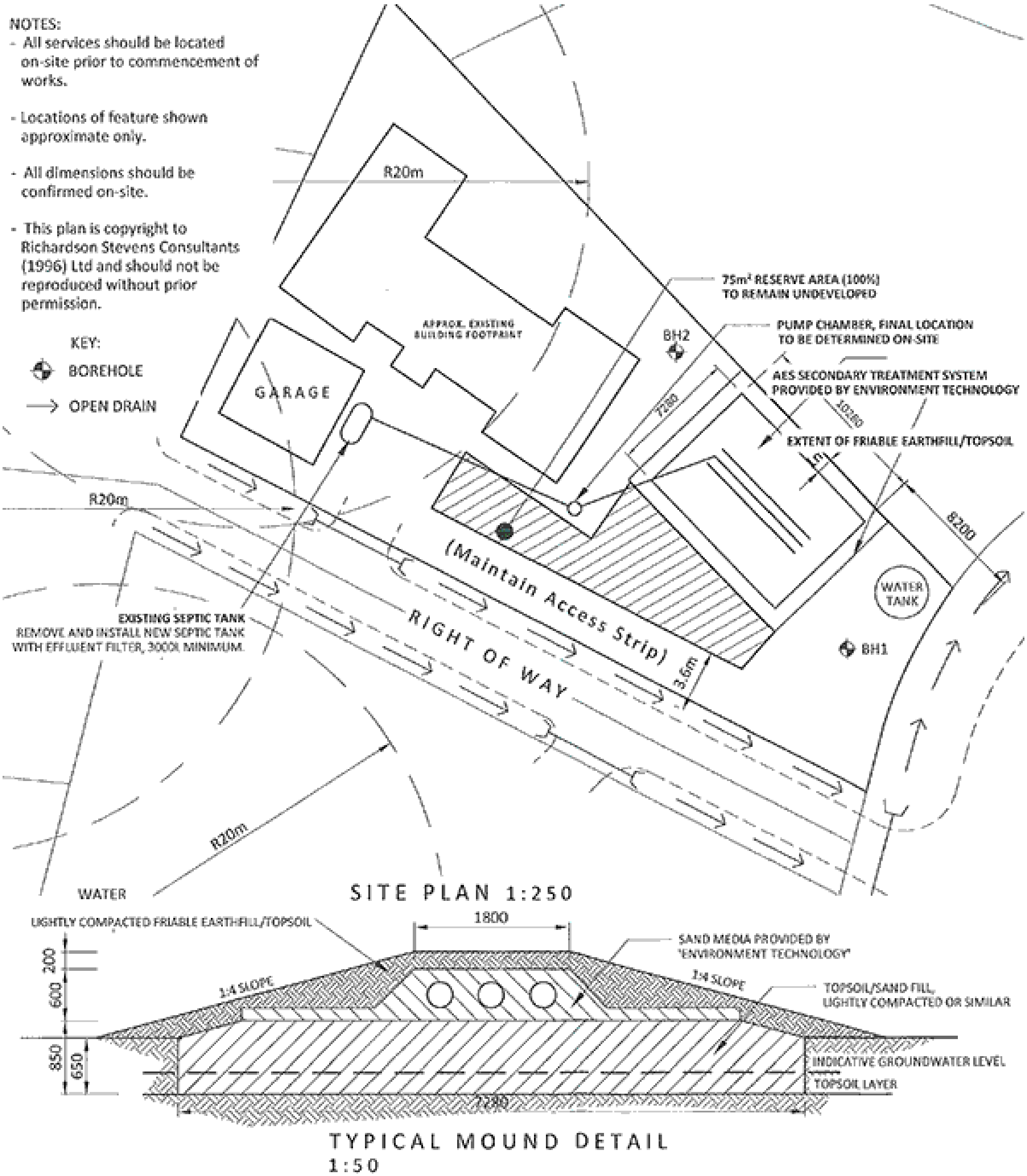
AES is a great solution for refitting a failing or non-compliant wastewater system at a bach, crib or holiday home. It's especially ideal in these situations because an AES system:

- Works well with intermittent use and shock loads
- Has no alarms to go off or pumps to fail
- Requires very little maintenance
- Can incorporate an existing septic tank in a new design
- Treats effluent to Advanced Secondary level - is easy on the natural (including human) environment

After installation and planting, a raised bed system can be seen above between the house and water tank.

The wastewater treatment system at a 3 bedroom holiday home with rainwater storage tanks on a flat 900m2 section near the water was required by the Regional Council to be upgraded. The existing primary treatment system comprising a septic tank and a single soakage trench had an overflow to a roadside drain, thereby having an effect beyond the property boundary. The soils are Category 5 clay and sand beneath topsoil. Groundwater was located within 0.6m and surface water within 15m from the disposal field requiring a resource consent. Soil category and site constraints meant a Secondary Treatment System was needed.

The owners asked for an AES system which was designed by Matthew Hargood of Richardson Stevens Consulting Engineers and installed by Russ Louie of Bay Plumbing Services. Designed for 725 litres per day (5 people using 145L/day) and using standard water reduction fixtures, the AES pipes were installed in a 75m2 base area raised bed to achieve 0.6m clearance to ground water. Section planting helps reduce subsurface loading. The installers said it was by far the most suitable system for this situation being coastally sensitive, a substantial dwelling on a small site with a high water table and not easily accessible being up a right of way. The owners were very pleased with the outcome.



For more information about AES wastewater treatment systems visit the resources page on our website, watch the introductory videos, or call us on 0800 927 834 (0800 waste H2O).

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